Day 1 Study Notes

HTML Basics	Web Architecture	Semantic HTML	Forms & Data
15	3	1	INF
Concepts	Tech Layers	Day	Curiosity

Topic: Understanding How Websites Actually Work

1 What is HTML?

HTML (HyperText Markup Language) is the **structure** of a webpage. Think of building a website like constructing a human body — each technology plays a distinct role.

Technology	Analogy	Purpose
HTML	Skeleton	Structure and layout
CSS	Skin / Clothes	Visual appearance
JavaScript	Brain	Behaviour and interactivity

Key Concept
CSS and JavaScript cannot work without HTML. HTML is always the foundation.

2 Tags, Elements and Attributes

HTML is made of **elements**. Each element follows a consistent formula:

```
<h1>Hello World</h1>
```

Component	Example	Meaning
Opening Tag	<h1>	Start of element
Content	Hello World	What is displayed
Closing Tag	</h1>	End of element
Element (full)	Hello World	Complete building block

Attributes

Attributes give elements extra information. They live inside the opening tag.

```
<a href="https://youtube.com">YouTube</a>
```

Part	Value	Purpose
Attribute Name	href	Identifies what info is given
Attribute Value	https://youtube.com	The actual destination URL

Images

```
<img src="dog.jpg" alt="Cute Dog">
```

Attribute	Purpose
src	Path to the image file
alt	Fallback text and screen reader support

Section Summary

[+] Every HTML element = Opening Tag + Content + Closing Tag

[+] Attributes add information (href, src, alt) inside the opening tag

[+] The img tag is self-closing — it has no content or closing tag

3 Head vs Body

Every HTML page has the same outer skeleton. The **head** stores invisible information for browsers and search engines; the **body** holds everything users actually see.

```
<html>
<head>
<title>My Website</title>
<meta charset="UTF-8">
<link rel="stylesheet" href="style.css">
</head>

<body>
<h1>Welcome</h1>
<p>Hello World</p>
</body>
</html>
```

	head	body
Visible?	No	Yes
Contains	Title, meta, CSS links	Text, images, buttons
Purpose	Browser and SEO info	User-facing content

4 Metadata, UTF-8 and DOCTYPE

Metadata

Metadata = data about data. A photo file stores not just pixels but also date taken, camera model, and GPS location — those are metadata. The title tag describes the webpage itself:

```
<title>My Website</title>
```

UTF-8 — Character Encoding

Computers store text as numbers (bytes). UTF-8 is the rulebook that maps those bytes back to readable characters from virtually every language:

```
<meta charset="UTF-8">
```

Without UTF-8	With UTF-8
Malayalam text -> ????	Malayalam -> correct characters
Hindi text -> symbols	Hindi -> correct characters
Emojis may break	All characters render correctly

DOCTYPE

```
<!DOCTYPE html>
```

Important Note
DOCTYPE is not a tag — it is a declaration. It tells the browser: Use modern HTML5 rules for this page. Without it, browsers fall back to quirks mode which can break your layout.

Section Summary

[+] Head = metadata (invisible) | Body = content (visible)

[+] Always add meta charset='UTF-8' to support all languages including Malayalam

[+] Always start with for modern HTML5 standards

5 Semantic Thinking

Do not pick tags randomly. Ask: *What is the purpose of this content?* Then pick the tag that reflects that purpose. This helps browsers, screen readers, and search engines.

Content	Purpose	Correct Tag
Irfan's Tech Journey	Main page title	<h1>
About Me	Section heading	<h2>
Python, HTML, CSS	A list of items	+
A paragraph of text	Body copy	<p>
Navigation links	Site navigation	<nav>
Page footer info	Footer section	<footer>

Design Principle
Semantic HTML is about meaning, not appearance. Use h1 because it IS a top-level heading, not because it looks big.

6 Buttons vs Links

This is one of the most common mistakes beginners make. The rule is simple:

Element	Use For	Example
Link (a href)	Navigation — go somewhere	Visit GitHub, Open profile
Button	Actions — do something	Login, Delete, Save, Submit

```
<!-- Navigate somewhere -->
<a href="https://github.com">My GitHub</a>

<!-- Perform an action -->
<button>Login</button>
```

7 Forms and the Key-Value Concept

Forms collect information from users. Every login screen, signup page, and search bar is a form under the hood.

Components of a Form

Component	HTML Tag	Purpose
Heading	<h1>, <h2>	Names the form
Labels	<label>	Describes each field
Inputs	<input>	Where users type data
Submit Button	<button type=submit>	Sends the form

Why Key-Value Pairs Matter

The server receives raw bytes. Without labels, it cannot tell what "Irfan" means. The **name** attribute on inputs creates the key:

```
<!-- Without name: server receives "Irfan" ??? -->
<!-- With name: server receives username=Irfan -->

<input type="text" name="username">
<input type="password" name="password">
```

Sent to server	Server understands
Irfan	Unknown value — useless
username=Irfan	Irfan is the username — useful

Section Summary

[+] Forms collect data via inputs — each input needs a name attribute

[+] name='username' creates the KEY; what the user types is the VALUE

[+] This becomes username=Irfan, password=secret123 sent to the server

8 Browser to Server to Database Flow

This is the most important concept of Day 1. When you fill a form and click submit, an entire journey happens behind the scenes:

User enters data
↓
Browser
↓
Server (validates data)
↓
Database (stores data)
↓
Response back to Browser
↓
User sees result

What the Server Validates

Check	Example
Email format	Does it contain @ and a domain?
Existing account	Is this email already registered?
Password rules	Min length, special characters?
Empty fields	Did the user skip required fields?

9 Why HTML Alone Cannot Build Instagram

HTML is only the structure. A real application needs the full stack:

Feature	HTML Only	Full Stack Needed

Show a login form	Yes	Yes
Store users	No	Database required
Verify passwords	No	Server / Python
Manage followers	No	Backend logic
Create personalised feeds	No	Python + DB queries
Send notifications	No	Backend services
Handle 1M users	No	Servers + cloud

HTML + CSS + JavaScript + Python/Django + Database = Real Web App

DAY 1 REFLECTION

Before vs After Day 1

BEFORE	AFTER
Website = a page	Website = a system
HTML does everything	HTML is just the structure
Tags are random	Tags have semantic meaning
Forms just look nice	Forms send key-value data to servers
No idea what a server does	Server validates, processes, stores

"I learned that web development is not just creating pages. It is about understanding how information moves through an entire system."

Biggest Takeaway

The web is a communication system. Every click, form submission, and page load triggers a conversation between the browser, a server, and a database. HTML is how we design the interface for that conversation — but the real intelligence lives in Python, the database, and the logic we will build over the coming weeks.

What is Coming on Day 2

- CSS — giving our HTML structure a visual appearance
- The box model — how every element takes up space on the page
- Selectors — targeting specific elements with styles
- Building the first styled webpage from scratch

Keep Coding. Keep Curious. Day 2 Awaits.